

DOULAN, China

WMO: 528360

Lat: 36.2958N

Lon: 98.0964E

Elev: 10466

StdP: 9.93

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.2	3.2	-23.9	2.2	16.5	-20.9	2.6	17.9	17.0	22.3	15.8	23.7	4.3	140	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	78.4	52.3	75.0	50.6	72.2	49.1	55.8	71.3	54.0	68.4	52.5	65.8	5.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
50.8	82.1	60.1	49.1	76.9	59.0	47.4	71.9	57.8	28.7	71.2	27.3	68.5	26.1	65.7	63.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.1	12.2	-5.7	84.1	4.3	2.7	-8.8	86.1	-11.4	87.7	-13.8	89.2	-16.9	91.2		
(5)			-7.3	58.3	4.1	2.3	-10.2	60.0	-12.6	61.3	-14.9	62.6	-17.8	64.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.9	17.0	23.5	31.7	41.7	49.2	55.4	60.0	59.5	51.7	39.9	28.6	19.2		
	DBStd	16.17	6.72	7.36	6.87	6.71	5.72	5.25	5.73	5.28	5.54	6.00	6.68	5.97		
	HDD50	4649	1023	743	567	261	84	14	1	1	41	317	642	955		
	HDD65	9215	1488	1163	1031	699	490	291	177	187	400	777	1092	1420		
	CDD50	949	0	0	1	12	59	175	311	294	92	5	0	0		
	CDD65	39	0	0	0	0	0	2	21	15	1	0	0	0		
	CDH74	392	0	0	0	1	1	25	201	146	18	0	0	0		
	CDH80	48	0	0	0	0	0	1	29	17	1	0	0	0		
	WSAvg	5.5	5.4	5.9	6.4	6.5	6.1	5.3	5.0	4.9	4.8	5.1	5.5	5.5		
	PrecAvg	8.1	0.2	0.2	0.3	0.4	1.1	1.9	1.9	1.1	0.7	0.3	0.2	0.1		
(15) Precipitation	PrecMax	13.8	0.4	0.8	1.1	1.5	3.1	5.6	4.3	3.7	1.9	1.1	0.7	0.5		
	PrecMin	4.2	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.2	0.0	0.0	0.0	0.0		
	PrecStd	2.4	0.1	0.2	0.2	0.4	0.7	1.0	1.0	0.7	0.5	0.2	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	50.0	59.7	67.9	72.4	77.2	83.5	82.3	76.5	65.3	52.1	43.3		
		MCWB	26.8	30.9	36.0	41.1	44.1	51.1	54.7	54.6	51.9	41.1	32.4	27.3		
	2%	DB	36.2	44.6	52.2	62.9	68.4	73.3	79.2	78.0	70.6	59.1	47.2	37.3		
		MCWB	23.3	28.2	32.3	38.4	42.3	48.5	53.3	51.9	48.0	37.7	30.2	24.2		
	5%	DB	32.0	40.0	48.0	58.9	65.1	70.4	75.6	74.5	66.6	55.0	43.5	33.5		
		MCWB	21.1	25.9	30.2	36.8	41.0	46.5	51.2	50.6	45.7	35.8	28.2	21.9		
	10%	DB	28.2	35.7	44.1	55.0	61.7	67.6	72.4	71.4	63.1	51.5	40.0	29.9		
		MCWB	18.7	23.7	28.4	34.9	39.7	45.3	50.3	49.1	43.5	34.3	26.5	20.0		
	0.4%	WB	27.0	31.1	36.3	42.1	46.8	55.5	58.5	58.2	54.8	43.9	33.1	27.7		
		MCDB	41.2	48.4	57.9	63.1	62.5	71.0	76.0	73.8	69.6	59.9	50.5	42.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	23.7	28.4	33.0	39.5	44.8	52.2	56.3	55.5	51.4	40.0	30.5	24.3		
		MCDB	35.3	43.4	50.4	59.8	60.2	65.5	72.2	71.1	63.9	54.1	45.8	36.7		
	5%	WB	21.2	26.2	30.8	37.6	43.2	50.1	54.4	53.9	49.0	37.5	28.6	22.1		
		MCDB	31.5	39.3	46.1	56.6	58.8	62.5	68.8	68.4	61.6	52.0	42.5	33.0		
	10%	WB	19.0	23.9	28.8	35.7	41.7	48.4	52.8	52.2	46.6	35.3	26.8	20.1		
		MCDB	27.9	35.1	42.9	53.2	57.0	60.9	66.2	65.4	59.0	49.6	39.0	29.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	19.4	19.2	20.2	19.6	18.7	18.7	20.0	20.2	19.9	19.4	19.5		
		MCDBR	23.2	24.3	24.3	25.2	25.3	24.0	24.2	24.7	24.7	24.0	22.7	22.9		
		MCWBR	15.0	14.2	12.4	11.2	10.2	8.7	8.7	8.6	9.6	10.9	12.2	14.4		
		MCDBR	22.7	23.1	22.6	23.2	20.0	18.5	19.2	19.8	20.2	21.0	21.8	22.3		
	5% WB	MCWBR	14.7	13.6	11.7	10.5	8.3	7.6	7.7	7.4	8.3	9.9	11.8	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.219	0.257	0.311	0.360	0.371	0.362	0.350	0.340	0.315	0.274	0.233	0.214		
	taud		2.477	2.367	2.173	2.042	2.099	2.229	2.346	2.360	2.399	2.432	2.470	2.499		
	Ebn at Noon		320	315	303	292	289	290	293	295	298	305	311	317		
	Edh at Noon		25	32	43	53	51	44	39	38	35	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		991	1245	1666	2038	2148	2080	2089	1980	1690	1459	1121	934		
	RadStd		72	73	98	94	123	156	146	140	117	65	60	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page